

OUAHID OMAR

AI-Focused Data Engineer

@ omarouahid.98@gmail.com
Casablanca, Morocco

+212680364513

in omar-ouahid

PROFESSIONAL SUMMARY

Experienced AI-focused Data Engineer with proven expertise in designing and implementing scalable data pipelines, machine learning infrastructure, and AI-powered data processing systems. Skilled in architecting end-to-end data solutions that support ML/AI workflows using modern cloud platforms and orchestration tools. Demonstrated ability to transform raw data into valuable business insights through robust ETL/ELT processes, feature engineering, and automated ML pipelines. Strong background in optimizing data systems for AI/ML workloads, performance, and scalability.

PROFESSIONAL EXPERIENCE

Data Engineer at DXC Technology

May 2024 - Present

Casablanca, Morocco

- Architected and implemented scalable data pipeline systems using AWS services (Fargate, ECR, Step Functions) with automated CI/CD deployment processes.
- Designed and built ETL pipelines for healthcare laboratory data processing, handling multiple data formats and ensuring data quality and consistency.
- Developed API-driven data ingestion systems that dynamically generate data workflows from OpenAPI specifications, enabling flexible data source integration.
- Implemented real-time data processing solutions using AWS Lambda and Step Functions for complex workflow orchestration and monitoring.
- Built and maintained data preparation workflows optimized for downstream analytics and machine learning applications.
- Established data governance and quality assurance processes, ensuring reliable data delivery across multiple business units.

Data Engineer at Berexia (External at SCOR)

Sep 2021 - Jan 2024

Casablanca, Morocco

- Designed and implemented automated data extraction and transformation pipelines using Python, reducing manual processing time by 70%.
- Built scalable ETL systems for insurance data processing, handling millions of records with high availability and fault tolerance.
- Developed data validation and quality monitoring frameworks to ensure data integrity across multiple source systems.
- Created RESTful APIs and microservices architecture using Flask for efficient data access and integration.
- Implemented real-time data visualization dashboards using Dash, enabling stakeholders to monitor key business metrics.
- Optimized database performance and designed data models for improved query efficiency and storage optimization.

Data Scientist at Adwebmaroc

Mar 2021 - Aug 2021

Casablanca, Morocco

- Developed an automated brand detection system with Python, OpenCV, TensorFlow, and YOLOv5.
- Used advanced image processing for brand element detection.
- Implemented OCR to enhance data richness and accuracy.
- Applied NLP for comprehensive analysis of extracted text.
- Enhanced system efficiency with a custom-tuned YOLOv5 model.
- Combined computer vision, OCR, and NLP for robust brand detection.

EDUCATION

Master's Degree in Engineering (Diplôme d'Ingénieur) - Data Science and Computer Science Engineering

Institut National des Postes et Télécommunications

📅 2018 - 2021

Intensive preparatory coursework in scientific disciplines

CPGE - Classes Préparatoires aux Grandes Écoles

📅 2016 - 2018

PROJECTS

Real-time Customer Data Pipeline

📅 2024

- Architected and implemented a comprehensive data pipeline solution for Catawiki's customer support system, processing real-time user interaction data.
- Built scalable data ingestion and transformation pipelines using Python and cloud-native services to handle high-volume customer data streams.
- Implemented vector database solutions (ChromaDB) for efficient storage and retrieval of customer interaction patterns and support documentation.
- Designed automated data quality monitoring and validation systems to ensure data integrity across the entire pipeline.
- Created RESTful API endpoints using Flask for real-time data access and integration with downstream applications.
- Established monitoring and alerting systems for pipeline performance optimization and proactive issue resolution.

SKILLS

Programming Languages

- Proficient: Python, SQL
- Familiar: Java, Scala, R

Data Engineering Platforms & Tools

- Big Data Processing: Apache Spark, Hadoop ecosystem
- Workflow Orchestration: Apache Airflow, AWS Step Functions
- Data Storage: PostgreSQL, MySQL, MongoDB, Redis
- Message Queues: Apache Kafka, AWS SQS, RabbitMQ
- Data Warehousing: Amazon Redshift, Google BigQuery, Snowflake
- Vector Databases: Pinecone, ChromaDB, Weaviate

Cloud Computing & Infrastructure

- AWS Services: Lambda, EC2, Fargate, Step Functions, CloudWatch, S3, RDS, Redshift, Glue, Kinesis
- Containerization: Docker, Kubernetes
- Infrastructure as Code: Terraform, CloudFormation
- CI/CD Pipelines: Jenkins, GitLab CI, AWS CodePipeline

AI/ML Data Engineering

- ML Pipeline Development: TensorFlow, PyTorch, scikit-learn, MLflow
- Feature Engineering: Advanced feature stores and ML data preparation
- Model Serving: Real-time and batch inference pipelines
- MLOps: Model versioning, monitoring, and automated retraining pipelines
- LLM Infrastructure: RAG systems, vector databases, prompt engineering workflows

ETL/ELT & Data Processing

- Data Manipulation: Pandas, NumPy, Dask
- Stream Processing: Apache Kafka, AWS Kinesis
- Batch Processing: Apache Spark, AWS Glue
- Data Quality: Great Expectations, custom validation frameworks

API Development & Integration

- RESTful APIs: Flask, Django, FastAPI
- Microservices Architecture: Design and implementation of scalable distributed systems
- API Gateway: AWS API Gateway, Kong

Data Visualization & Monitoring

- Dashboards: Plotly, Dash, Grafana
- Monitoring: CloudWatch, Prometheus, DataDog
- Business Intelligence: Tableau, Power BI

Version Control & Collaboration

- Git, GitHub, GitLab
- Agile methodologies and DevOps practices

LICENSES & CERTIFICATIONS

- AWS Certified Solutions Architect (In Progress)
- Amazon SageMaker - LinkedIn
- Google Cloud Platform Data Engineering (In Progress)